Entertainment Computing 55 (2025) 101049

Contents lists available at ScienceDirect

Entertainment Computing

journal homepage: www.elsevier.com/locate/entcom

Lock the look: Recommending trendy looks for fashion products using natural language processing

Manjarini Mallik [c,a], Tushti Thakur [b], Chandreyee Chowdhury [a,*]

[a] Computer Science and Engineering Department, Jadavpur University, Kolkata, India
[b] Mechanical Engineering Department, Jadavpur University, Kolkata, India
[c] CSE(AIML),CSBS Department, Institute of Engineering & Management, Kolkata, India

ARTICLE INFO

Keywords:
Natural language processing
Fashion recommendation system
Bag-of-words
Text similarity analysis
E-commerce

ABSTRACT

The recreation of looks established by favorite movie characters or fashion icons is a popular trend in this decade. It is difficult to find out the dresses and accessories required to develop that look as current product recommendations are mostly based on history of users' choices. There exists computer vision-based solutions that check image-wise similarities between the desired looks and available fashion products from e-commerce stores. However, this is a resource hungry complex process as plenty of product images would be analyzed. In this work an NLP-based lightweight look recommendation system is proposed. In the proposed approach, multiple text descriptions of trendy looks are collected from different websites to build the training dataset. A subset of two benchmark datasets (Myntra Products Dataset and Ajio Products Dataset) have been used for recommendation. Using the bag of words technique, text datasets are embedded, and a set of looks is recommended for each product. The system is validated using Cosine similarity and Cohen's kappa metrics. Products in the test dataset have been mapped to their 1st and 2nd highest recommended looks with positive scores. We observed a minimum score of 0.6 and 0.2 for Cosine similarity and Cohen's kappa respectively, representing appreciable performance.

1. Introduction

The widely developing field of e-commerce is continuously revolutionizing the fashion industry [1]. It has amplified the demand and provides a number of choices to the customer, from where they can easily select the products as per their choice. Consumers expect seamless shopping with better-personalized recommendations that fulfill their requirements. Also, there is an abundance of options that can lead to an overload of information, making it challenging for shoppers to discover products that truly resonate with their immediate demands, tastes, and preferences.

Most online shopping platforms host a variety of fashion products under different categories. Whenever the user search for some particular item, it generates the Product Display Page (PDP), which showcases the relevant product items. Machine learning based recommendation techniques, such as collaborative filtering and content-based filtering are used for recommending products based on the users' search history, preferences, and product attributes [2,3]. However, these techniques often do not perform well in capturing the dynamic complexities such as changes in trend, personal preferences, etc. Also, these processes rely on broader patterns rather than deep individual insights, and struggle to break down the fine detailed information inherent in user's query. Investigating hidden information can be the basic requirement for some cases, such as, to build a desired look as per customer's wish.

Recreating the look of favorite movie characters, models, or fashion influencers is in high demand in the current era [4]. There are plenty of posts in social media on look recreations having numerous views and likes. Sometimes, links of different products are provided with their posts that have been used to recreate the look, so that viewers can buy. This phenomenon represents a strong connection with the e-commerce. This fact motivated us to approach the same matter from a different perspective. How would it be if customers get varieties of products on the e-commerce site, linked with some trendy look?

Some researchers recently mapped this query with computer vision domain [5,6]. In computer vision based fashion product recommendation approach, desired look is provided with an input query image, then it analyses with deep learning models and required products are identified to develop the look. Often reference articles or blogs are also investigated for more referral images for the look. This process allows

* Correspondence to: Computer Science and Engineering Department, Jadavpur University, Kolkata 700032, India.
E-mail addresses: manjarini.mallik@gmail.com (Manjarini M.), tushtithakur1234@gmail.com (Tushti T.), chandreyee.chowdhury@jadavpuruniversity.in (Chandreyee C.).

https://doi.org/10.1016/j.entcom.2025.101049
Received 7 January 2025; Received in revised form 25 October 2025; Accepted 30 October 2025
Available online 1 November 2025

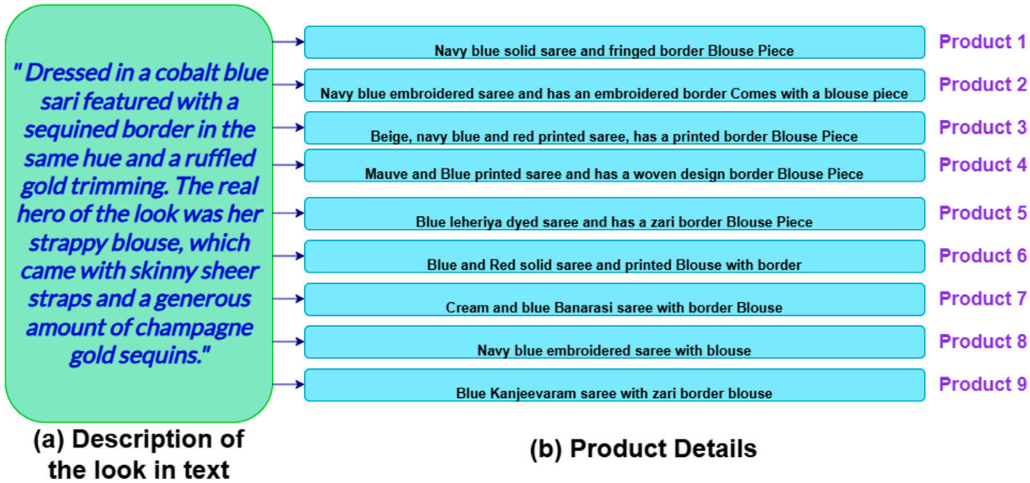

Fig. 1. A representation of the look recommendation system — (a) Text description taken from a web source representing the desired look, (b) Sentences representing different products taken from the Myntra product dataset.

the recommendation system to be independent of the user's previous purchase history or web visits. However, feature abstraction, segmentation or classification of huge number of images is a resource-intensive and data hungry process which becomes more heavier when images from different sources contain complex backgrounds. As an alternative, researchers proposed deep learning models to abstract crucial features of products in detail with text representations [7]. Now, how these detailed text features can be utilized to develop a recommendation system without depending on the user's previous purchase history remains still unaddressed.

To address this challenge, we have chosen to employ natural language processing (NLP) instead of computer vision. The escalating usage of computational resources in the modern era necessitates the use of multiple task planning algorithms and other techniques to maximize system efficiency [8]. Thus, the need for a lightweight system has become inevitable. To keep the recommendation system lightweight, no large language model (LLM) is used. Instead, Bag of Words (BoW) method is integrated with supervised learning algorithms like Random forest and Artificial neural network. The idea is, on the users' side, the e-commerce portal will provide some popular looks as options. Users can select their desired look, and products required to build the look would be recommended from the store. The recommendation system which is required for this service's back end is proposed in this work. As shown in Fig. 1, each look can be mapped with one or more products based on their similarities. The companies can also indicate the popular looks that can be built with a product. To recommend one or more looks for each product, the similarity between the textual descriptions of the products and the looks would be analyzed and machine learning techniques would be incorporated as proposed in this work.

In the proposed approach, multiple textual descriptions for a single look are collected from different fashion or movie blogs which form the training dataset. Then, the text features are extracted from those descriptions and embedded to numeric feature vectors using Bag of Words (BoW). On the other hand, the product database containing product descriptions acts as test data. Finally, ML classifiers are trained with the movie looks dataset, and recommend from the product datasets to predict the set of looks to which each product can be linked. An example is shown in Fig. 2, representing the mapping between a desired look and the products which have been assigned with that look according to the proposed approach.

Proposed approach is tested on two benchmark datasets — Myntra and Ajio fashion product dataset. A subset from each dataset was selected for experimentation based on the gender, dress and prime color. The training dataset contains data collected from various websites on the different looks of well-known models and celebrities. To

Fig. 2. An example of the mapping of the desired look to related products.

the best of our knowledge, this is the first ever text-based fashion look recommendation approach. Accordingly, contributions of the paper are as follows -

- A resource-friendly recommendation system is proposed that recommends popular looks for each product of an e-commerce product dataset.
- The proposed system is trained on fashion and movie blogs while tested on e-commerce datasets such as, Myntra and Ajio product dataset.
- Performance of the recommendation system has been evaluated with Cosine similarity and Cohen's kappa metrics.

Rest of the paper is structured as follows. Section 2 shows related works on recommendation systems. In Section 3, the methodology has been discussed, which is followed by the discussion on results as in Section 4, and the conclusion in Section 5 respectively.

2. Related work

The fourth industrial revolution has emphasized data-driven decision-making in many industries [9] such as manufacturing, healthcare, transportation, e-commerce etc, for simplifying their applications. The recommendation systems in fashion e-commerce platforms have evolved significantly over the recent years. To make a recommendation system less dependent on human experiences and knowledge expertise, AutoML has been introduced to automatically understand crucial features and recommend accordingly [10]. In this section, some recently proposed approaches are discussed for fashion recommendation system.

Collaborative filtering (CF) is a widely used method in recommendation systems. In [1], researchers proposed a CF-based algorithm for recommending products, prioritizing seasonal demands of customers and analyzing the changes in preferences over time among previously bought items. In [2], collaborative filtering was integrated with deep learning and web usage mining to recommend products on image datasets. However, the system uses large convolutional neural network (CNN) based models for feature extraction which makes the process resource-hungry. Capturing users' likes and dislikes is the main issue of a CF-based recommendation system, which often has trouble with new fashion items or abrupt changes. An alternate approach is Content-based filtering (CBF), where product items are separated into categories considering features like feedback and keywords to recommend similar items. In [3], CBF was used on apparel data along with CNN for image-based product recommendations, while [11] focused on individual user preferences and fashion trends. In [12], fusion of CBF and supervised classifiers are used to develop recommendation system. CBF was used to filter the products based on user's preference. Then, among all recommended products the one with the highest rating is identified. Though the strategy is different from CF, CBF is also dependent on previous purchase and user interaction data, limiting the performance of the system for something completely new.

An approach for proposing an outfit similar to some desired product was proposed in [13], where the model was meant for generating fashion outfits using self-attention mechanisms and Bi-LSTM networks. The goal was to generate an useful training dataset analyzing user's clickstream activity. But, what if the user does not click any page to find their desired product? Breaking the chain, in [5], a different approach was proposed to recommend similar products to build user's desired look. Users ask for their favorite fashion look with some query image. The conventional idea of applying CNN to fashion images [6,14,15] was followed in order to extract the key features from the query image. Similar images from various reputed fashion articles were also investigated to find out related products from the e-commerce product catalog image database. However, the predictable limitation still holds: feature abstraction from images and mapping those abstracted features on the product dataset. Researchers in [7] proposed a transfer-learning based model to generate detailed textual captions for each apparel image. CNN and Long Short-Term Memory (LSTM) both were used to build the image caption network. This work not only provides finely annotated information about the fashion product to users but also opens a path for a text-based fashion recommendation system.

Textual data has contributed in the product recommendation system from a different perspective. Analyzing user's sentiment from their feedback on previously purchased product can help to build an useful product recommendation system [16,17]. However, it is difficult to get feedback for all kinds of sentiments, as people usually express complaints more often than satisfaction. Even sometimes the detailing of the product is missing, due to which it is liked or disliked.

From the existing works, it is clear that the recommendation system is stepping out of the dependency of previous purchase history or web visits. Emerging approaches are focusing on recommending fashion products that resonate with user preferences, mostly implemented on image datasets though. On the other hand, existing works made it feasible to represent products in detail with textual information. Integrating these two research directions, we have proposed a text-based fashion look recommendation system as presented in the following section.

3. Proposed methodology

The different execution steps of the proposed approach for recommending popular looks for the fashion products of e-commerce websites are presented in this section. The proposed framework is represented in Fig. 3. The detailed steps of the framework has been discussed as follows.

1. Data collection and feature extraction:

The goal is to train the recommendation system with detailed textual descriptions of different popular looks so that it becomes capable to link a product to the most similar look based on the product's textual description. To build the training dataset, we explored various websites that contain textual descriptions in sentences of some popular movie character looks. Multiple sentences for one single look were collected from different websites to strengthen the feature vectors and stored. Total 1000 samples are collected in the entire training dataset. Each movie character look is given a unique integer class label. After initial training data collection, from each sentence 24 features were extracted and stored as respective attributes of the dataset. The noun forms of dress, color, material, and design names are the feature values. One training data sample with all attribute values is shown in Fig. 4. The number of features may vary for different datasets. In our case, one single look consists of at most three dresses, and for each dress we choose to consider three prime features: material, color (prime color and others), and design, for which 24 features are found to be sufficient. Domain of words that can be assigned to a particular feature were previously stored in dictionaries. Fig. 5 represents the values stored in the domain for the feature Dress_1 Material_1. In this dataset mainly women clothes are considered, and the values are selected accordingly. As the diversity in gender and product types increases, the domain associated with each feature will also expand. Therefore, in a real-world system, the dictionary that defines the feature domains must be dynamically updated in accordance with the inclusion of a certain number of new looks in the training dataset. The feature information can be stored in a JSON file using a nested dictionary structure, where the top-level keys represent categories such as "Men", "Women", "Kids", and "Couple". Each of these keys maintains an independent feature set containing the relevant feature names and their corresponding domain values, enabling efficient retrieval and integration during model training. As these new looks are added, the model should undergo incremental or progressive learning [18], enabling it to adapt and acquire knowledge about emerging trends without forgetting previously learned information.

In cases where a particular feature was not available for some look, such as when the color of a dress was not explicitly mentioned in the look description on the website, the missing value is filled with a 'nan'. For testing, two benchmark fashion product datasets are used — Myntra[1] and Ajio[2] product dataset. Since the training dataset contains 1000 samples, to maintain an 80-20 train-test split ratio, from each of the Myntra and Ajio datasets, 250 samples were chosen based on the gender and prime dresses. Two separate testing datasets were created in the same format, in order to be evaluated separately. Each product was assigned to a textual description in one or more sentence(s), from which 18 key features were extracted in a similar way as in the training dataset. This is summarized as data preprocessing block of Fig. 3.

2. Text embedding using Bag of Words:

The train and test datasets store extracted features in text format. To feed these features as input to a machine learning model, they are required to be converted into numeric values. Using Natural Language Toolkit (NLTK), each feature is converted to a token containing only lowercase letters. Subsequently, a Bag of Words model was created, and also the top 100 most frequent words were identified using a frequency-based approach. Following this, each sentence is represented as a binary vector, where each element indicates the presence or absence

[1] https://www.kaggle.com/datasets/shivamb/fashion-clothing-products-catalog

[2] https://www.kaggle.com/code/swetanishad/ajio-fashion-dataset-eda/input

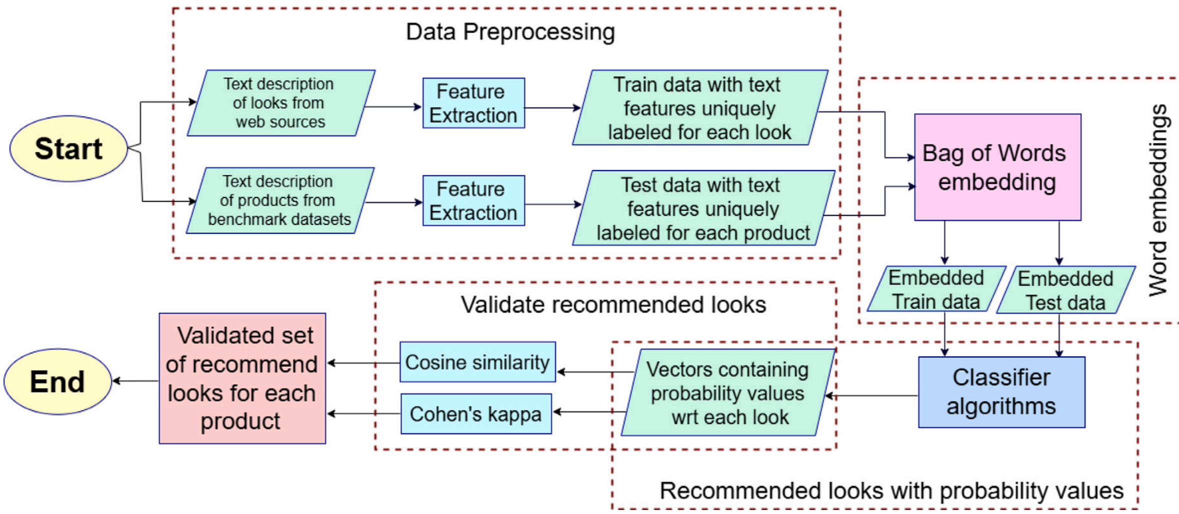

Fig. 3. Proposed framework for recommending similar looks for fashion products of e-commerce websites.

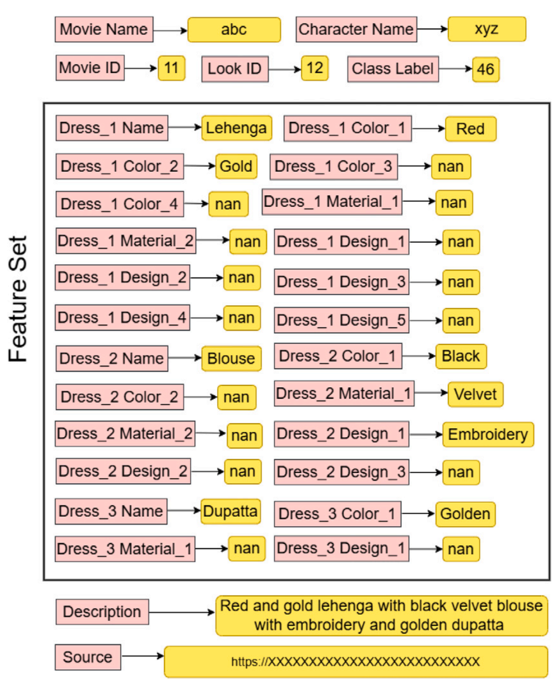

Fig. 4. Attributes and their values for one sample in the training dataset.

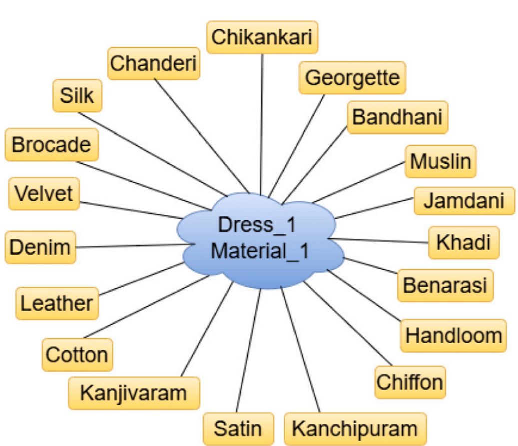

Fig. 5. Domain of feature values for Dress_1 Material_1.

of a frequent word. The number of most frequent words may vary for different datasets depending on the diversity of features. This process resulted in the creation of a feature matrix that was used for training and testing the machine learning classifiers.

3. Look recommendation for products:

Two different kinds of classifiers-Random Forest (RF) and Artificial Neural Network (ANN) have been employed to classify the data. Any supervised learning classifier that does not employ lazy learning could have been utilized though. The two classifiers RF and ANN are chosen as their learning process are completely distinct. RF follows bootstrapping and feature randomness, whereas ANN follows Gradient descent optimization to learn continuous, nonlinear relationships in data, making it effective for complex patterns.

When the samples from the benchmark product datasets are fed to the trained model for classification, based on their learning, they return a vector with each entry in the vector representing the probability of the product's belonging-ness to a specific look (class). Sorting this probabilities, we can rank the looks recommended for each product. For example, the class with highest probability score represents the 1st recommended look, then the class with second highest probability score represents the 2nd recommended look and so on. Analyzing this probability vector, for each product, a set of recommended looks can be decided for which the products can be a part. This is shown in Fig. 3.

The x_train data comprised the feature values of the movie looks, x_test data comprised the feature values of the products from the benchmark datasets. The y_train contained the class labels representing unique looks. As the test data samples, i.e., the products from the e-commerce website, are not already assigned to any look, there is no y_test in this experiment. For this reason, there is no scope to compute classification accuracy. So, the challenge remains in validating the model's predictions. For this purpose, the probability vector returned for each test sample (representing a unique product) has been investigated and validated through checking similarities between the text features of the looks and the product, explained as follows.

4. Validation of look recommendation:

After applying the classifiers, the similarities between each product (from product dataset) and its recommended looks (from training data) are validated using two metrics — Cosine similarity [19] and Cohen's Kappa method [20]. Cosine similarity is a common method in natural language processing used to compare multiple word vectors and analyzing semantic similarities among texts. In this work, the

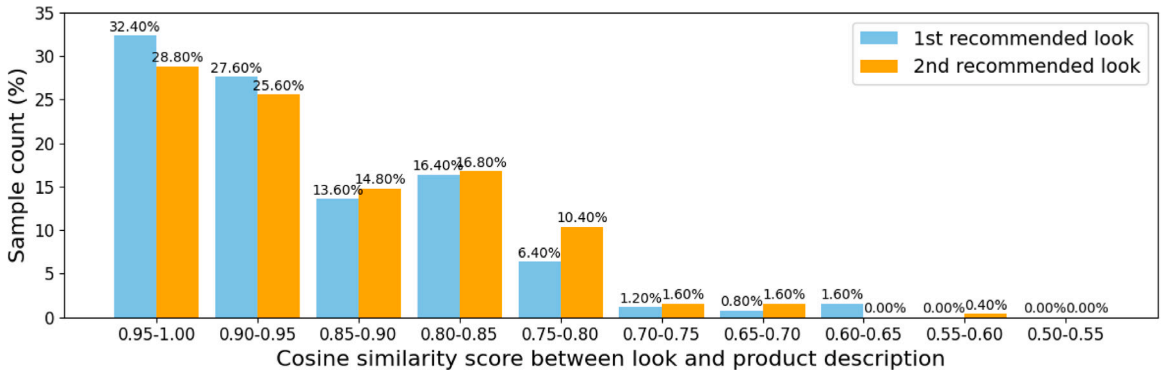

Fig. 6. Similarity analysis between Myntra dataset products and its recommended looks using Random Forest classifier and Cosine Similarity method.

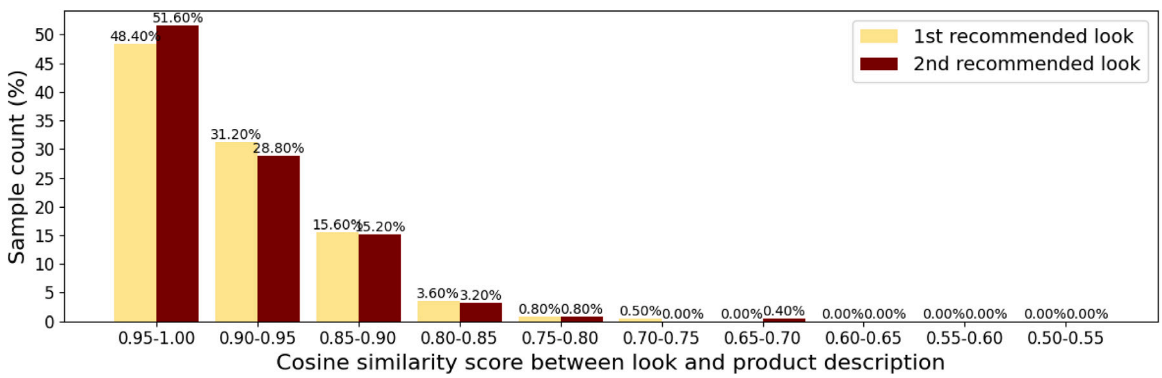

Fig. 7. Similarity analysis between Myntra dataset products and its recommended looks using Support Vector Machine (SVM) classifier and Cosine Similarity method.

cosine similarity is computed using spaCy [21]. The formula for cosine similarity between two word vectors A, B is presented in Eq. (1).

$$\frac{A \cdot B}{\|A\| \ \|B\|} \tag{1}$$

Another metric used is Cohen's kappa score that is used to quantify the level of agreement or consistency between two numeric vectors. Both metrics' scores range between −1 and 1, where 1 indicates very strong agreement, 0 indicates neutral agreement and −1 indicates opposing agreement. In this work, the text sentences are embedded using Bag of words method before computing Cohen's Kappa score. The formula is presented in Eq. (2) where P_o is observed agreement and P_e is expected agreement.

$$\frac{P_o - P_e}{1 - P_e} \tag{2}$$

4. Experimental results

Proposed look recommendation approach has been experimented on two benchmark datasets — Myntra product dataset and Ajio product dataset. The movie looks data are fed as training input data to three classifiers — Random Forest (RF), Support Vector Machine (SVM) and Artificial Neural Network (ANN). The ANN uses MLPClassifier from scikit-learn Python library, with one hidden layer of 100 neurons. The model applies a feedforward architecture trained via backpropagation. It runs up to 1000 iterations to minimize the loss and learn the mapping from input features to output classes. The results are presented as follows.

In Fig. 6, the result of RF classifier on Myntra product dataset is analyzed with cosine similarity score. The looks with the highest and second-highest probabilities are only considered to make the visualization neat and non-clumsy. Cosine similarity score can range between −1 to +1, where a positive score indicates strong similarity, a negative score indicates opposing characteristics and zero presents neutrality. For the 1st recommended look it is observed that, 32.40% of the products are mapped with the look with a cosine similarity score in the range $0.95 - 1.0$, 27.60% with a score greater $0.9 - 0.95$ and so on. All of the products have mapped with their 1st recommended look with a cosine similarity score greater than 0.60. This analysis indicates that for each product the 1st recommended look's textual description is significantly similar to the product's textual description. When the 2nd recommended look is considered, comparatively lower number of products are mapped with greater similarity score. This shows the effectiveness of the classifiers in predicting the most appropriate product with the highest probability. Interestingly, the minimum similarity score lies in the range $0.55 - 0.60$ indicating notable similarities between the products with their 2nd recommended looks as well. When SVM is used to analyze the cosine similarity scores, it is observed that in Fig. 7 for both 1st and 2nd recommended looks, around 50% of the products are mapped with the look with a cosine similarity score in the range $0.95 - 1.0$, and the minimum similarity score lies in the range $0.70 - 0.75$. This significantly reflects the effectiveness of the proposed recommendation system.

The next experiment represents the analysis of Cohen's kappa score on the same dataset using ANN and the results are shown in Fig. 8. This metric is applied on numeric data. Thus, this score is computed based on the word vectors from the train and test datasets. Around 14% of the 1st recommended looks are mapped to products with an almost perfect (0.8–1.0) score while 42.80% are mapped with substantial score of (0.6–0.8). Rest 33.20% are mapped to moderate and 10% to fair strength [22]. However, greater number of products are mapped with their 2nd looks with lower strengths thus maintaining parity with cosine's similarity. Fig. 2 indicates the corresponding image look of a test sample.

Experimentation has been conducted on another product dataset, the Ajio product dataset to validate the generalizability of the results.

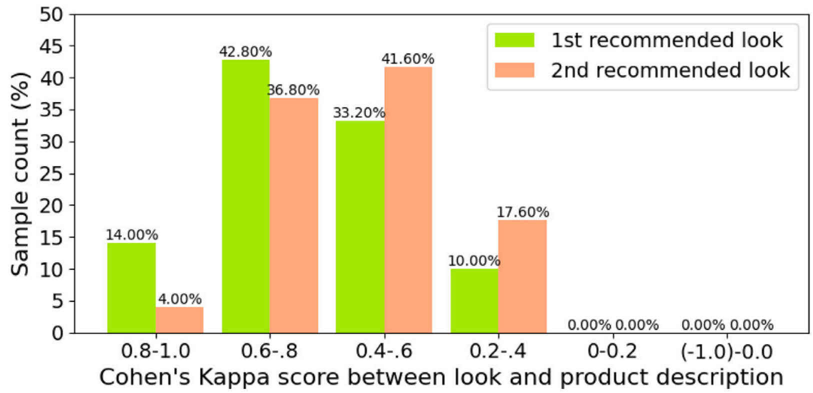

Fig. 8. Similarity analysis between Myntra dataset products and its recommended looks using Neural Network classifier and Cohen's kappa method.

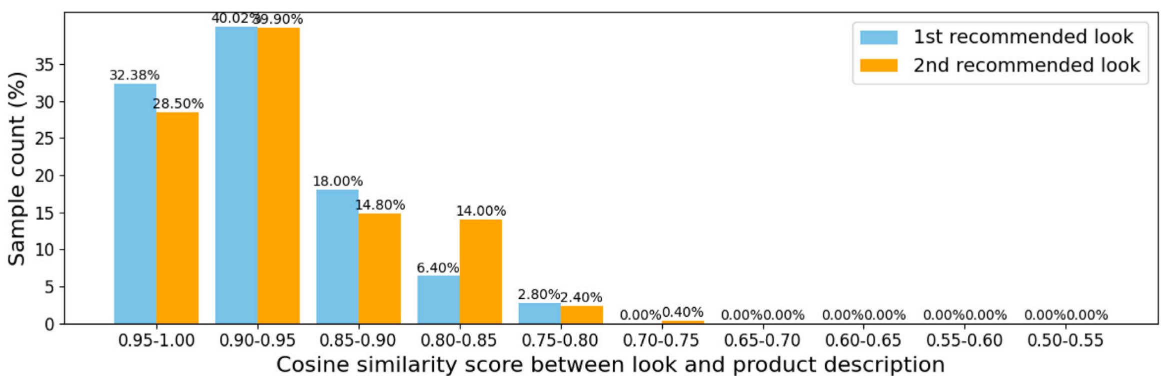

Fig. 9. Similarity analysis between Ajio dataset products and its recommended looks using Random Forest classifier and Cosine Similarity method.

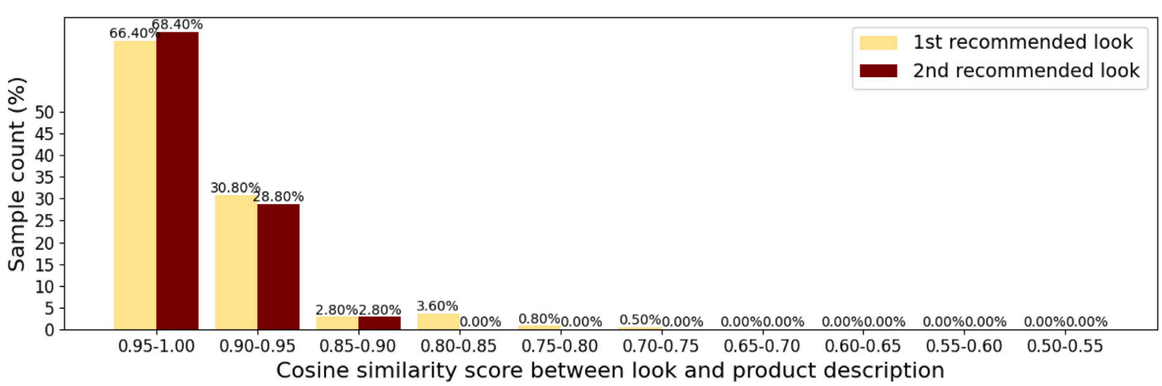

Fig. 10. Similarity analysis between Ajio dataset products and its recommended looks using SVM classifier and Cosine Similarity method.

The Cosine similarity results obtained using RF and SVM have been represented in Figs. 9 and 10 respectively, while Fig. 11 represents the Cohen's kappa score. Cosine similarity score is found impressive for this dataset; almost all products being mapped with a score greater than or equal to 0.75 for both 1st and 2nd recommended looks using both RF and SVM. Cohen's kappa score lies within similar range like that of the Myntra dataset, majority of products recommend 1st and 2nd looks with score greater than 0.4, indicating moderate to strong similarity. It shows that the Ajio dataset better correlates with the looks however, the system is able to recommend products matching with the trendy looks.

Model training with augmented training dataset: Since there is no benchmark dataset for different trendy looks, with much effort we collected 1000 training samples for our experiments as detailed in Section 3. To further increase the dataset, we have generated 1000 more data samples from the 1000 real training samples, using the text-to-text generation model T5_based, published on the Hugging Face platform,[3] designed to create paraphrased versions of a given text. In this way, the number of samples per look is also increased, which helps the supervised model to learn look-specific features more efficiently. Then, all 2000 samples have been used to train the supervised learning model. Figs. 12 and 13 represent the cosine similarity scores obtained using Random Forest, between highest recommended look and product

[3] https://huggingface.co/docs/transformers/model_doc/t5

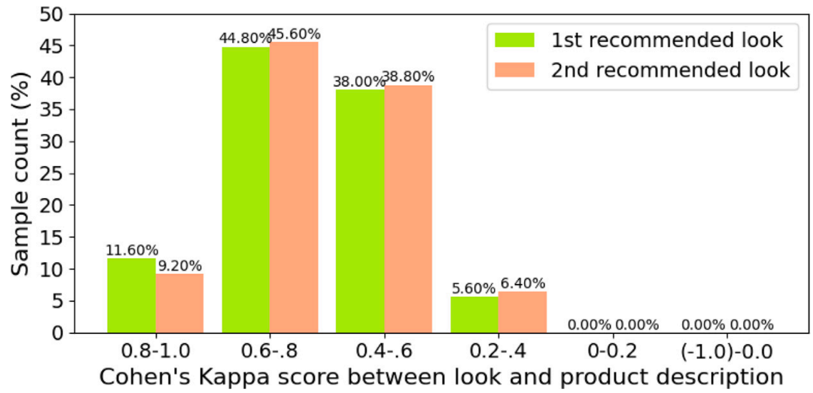

Fig. 11. Similarity analysis between Ajio dataset products and its recommended looks using Neural Network classifier and Cohen's kappa method.

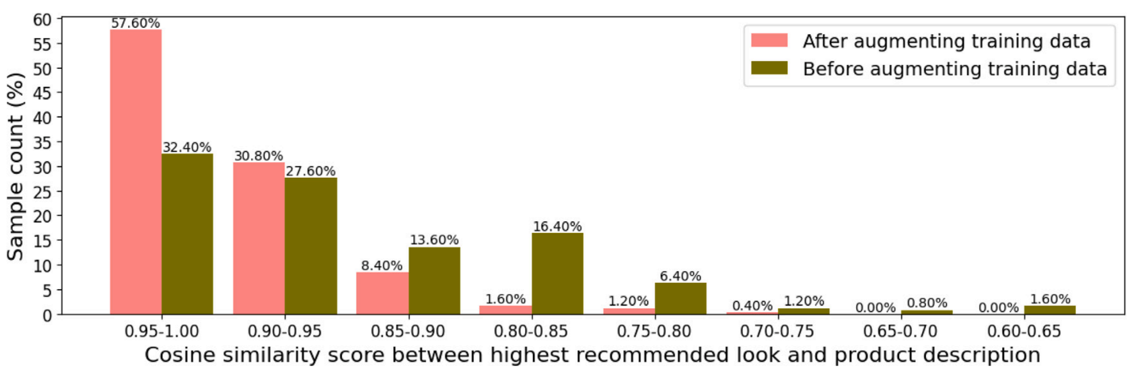

Fig. 12. Cosine Similarity Score analysis between Myntra dataset products and its highest recommended look using Random Forest, before and after training data augmentation.

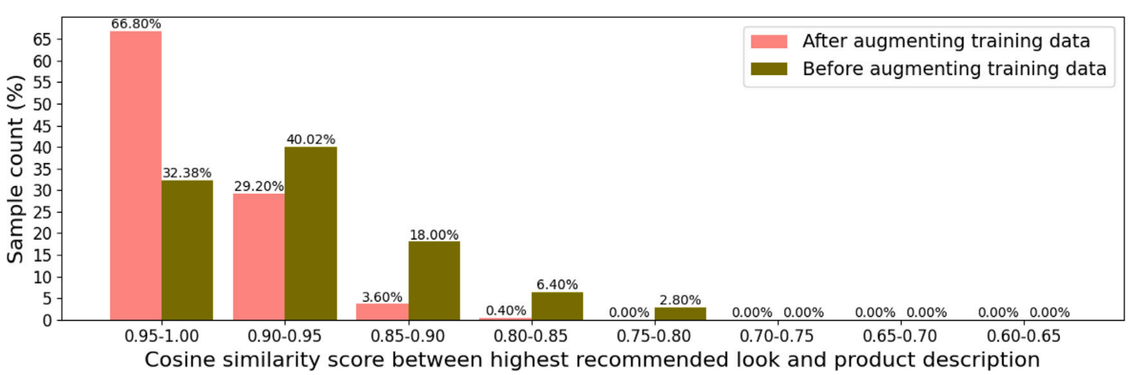

Fig. 13. Cosine Similarity Score analysis between Ajio dataset products and its highest recommended look using Random Forest, before and after training data augmentation.

description, when trained using 1000 samples before data augmentation, and with 2000 samples after data augmentation. We can see that, after increasing the number of training samples, greater percentage of recommendation could be obtained with higher similarity scores.

5. Conclusion

This paper presents a novel text-oriented lightweight approach to effectively recommend trendy looks for the fashion products of different e-commerce websites, which in turn can facilitate users to recreate their favorite looks. Textual descriptions of different trendy looks from various web sources have been collected to build a training dataset for looks. Two e-commerce product datasets from Myntra and Ajio were used for recommendation. For each of the product, a set of trendy looks was predicted using Random Forest and Artificial Neural Network classifiers. The prediction performance has been validated through the Cosine similarity and Cohen's kappa scores for the highest and second highest recommended looks. It is found that, for each product, its first recommended looks exhibit a better positive score for both metrics. The second recommended products also exhibit fair semantic similarity.

This indicates the effectiveness of product(s) recommendation depending on a trendy look as per users' choice rather than always going by the past purchase history of the users. The model is lightweight as compared to vision-based deep learning models for product recommendation. Thus, two lessons are learned from the work – (i) simple encodings like BOW can perform well, especially when the dataset is small or domain-specific. (ii) The limited amount of training data might restrict the superiority of the learning model, which is possible to overcome using generative models.

The work has a few limitations as well. Although the proposed system is found to be capable enough to recommend trendy looks, the real training dataset is small in size, and it mainly focuses on women's section data. So, we plan to build a larger text dataset of trendy looks for varieties of sections including men's, women's, kids, couple, and so on for better recommendation. The feature set will be increased as well in terms of variations and dimensionality. Thus, in future, we plan to automate the feature extraction process using latest NLP methods available in python (for example, pos_tag[4]). Furthermore, we could explore the possibility of combining the two notions-user's preferences with trendy looks. However, there is another matter of concern from an industrial perspective. The better the recommendation system is, the more likely it is to recommend products that closely resemble the real appearance of the character. In such cases, the issue of copyright must be addressed through proper rules and regulations, if applicable.

CRediT authorship contribution statement

Manjarini Mallik: Writing – original draft, Investigation, Conceptualization. **Tushti Thakur:** Writing – original draft, Visualization, Data curation. **Chandreyee Chowdhury:** Writing – review & editing, Supervision, Investigation, Formal analysis.

Declaration of competing interest

The authors have declared no conflict of interest.

Data availability

Data will be made available on request.

References

[1] Hyunwoo Hwangbo, Yang Sok Kim, Kyung Jin Cha, Recommendation system development for fashion retail e-commerce, Electron. Commer. Res. Appl. 28 (2018) 94–101.
[2] B. Bhavana, N. Raheja, DeNet_SVM: Product based recommendation system using deep learning and web usage mining, in: 2022 Seventh International Conference on Parallel, Distributed and Grid Computing, PDGC, 2022.
[3] R. Kumar, K. Jyotsna, K. Ganta, R.S. Nori, E-commerce recommender system using product data, Int. J. Sci. Technol. Res. (2020).
[4] Suk-Young Kim, Beauty and the waste: Fashioning idols and the ethics of recycling in Korean pop music videos, Fash. Theory 25 (1) (2021) 53–73.
[5] Abhinav Ravi, Sandeep Repakula, Ujjal Kr Dutta, Maulik Parmar, Buy me that look: An approach for recommending similar fashion products, in: 2021 IEEE 4th International Conference on Multimedia Information Processing and Retrieval, MIPR, 2021, pp. 97–103.
[6] Hessel Tuinhof, Clemens pirker and markus haltmeier, image-based fashion product recommendation with deep learning, in: International Conference on Machine Learning, Optimization, and Data Science, 2019, pp. 472–481.
[7] R.H. Banerjee, A. Ravi, U.K. Dutta, Attr2Style: A transfer learning approach for inferring fashion styles via apparel attributes, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2020, pp. 1–7.
[8] S. Menon, S.R. Addula, A. Parkavi, C. Subbalakshmi, V.B. Dhandayuthapani, K.S. Pokkuluri, A. Soni, Streamlining task planning systems for improved enactment in contemporary computing surroundings, SN Comput. Sci. 5 (8) (2024) 993.
[9] S.R. Addula, G.S. Sajja, Automated machine learning to streamline data-driven industrial application development, in: 2024 Second International Conference Computational and Characterization Techniques in Engineering & Sciences, IC3TES, IEEE, 2024, pp. 1–4.
[10] R. Zheng, L. Qu, B. Cui, Y. Shi, H. Yin, Automl for deep recommender systems: A survey, ACM Trans. Inf. Syst. 41 (4) (2023) 1–38.
[11] Pankaj Agarwal, Sreekanth Vempati, Sumit Borar, Personalizing similar product recommendations in fashion E-commerce, AI for fashion, in: The Third International Workshop on Fashion and KDD, 2018.
[12] Sariya Fathima, P. Bavithra Matharasi, Fashion fusion: Exploring apparel recommendation systems across companies using machine learning approach, Int. J. Inf. Technol. Res. Appl. 3 (2024).
[13] M. Madan, A. Chouragade, S. Vempati, The joy of dressing is an art: Outfit generation using self-attention bi-LSTM, in: Joint European Conference on Machine Learning and Knowledge Discovery in Databases, Springer, 2021, pp. 218–233.
[14] Betul Ay, Galip Aydin, Visual similarity-based fashion recommendation system, in: Generative Adversarial Networks for Image-To-Image Translation, 2021, pp. 185–203.
[15] Ssvr Kumar Addagarla, Anthoniraj Amalanathan, e-SimNet: a visual similar product recommender system for E-commerce, Indones. J. Electr. Eng. Comput. Sci. 22 (2021) 563–570.
[16] Ruchira Muchhal, Mohit Jain, Implementation of product recommendation model using sentiment analysis, Int. J. Res. Anal. Rev. 5 (2018).
[17] L. Yang, Y. Li, J. Wang, R.S. Sherratt, Sentiment analysis for E-commerce product reviews in Chinese based on sentiment lexicon and deep learning, IEEE Access 8 (2020) 23522–23530.
[18] G.M. Van de Ven, T. Tuytelaars, A.S. Tolias, Three types of incremental learning, Nat. Mach. Intell. 4 (12) (2022) 1185–1197.
[19] F. Rahutomo, T. Kitasuka, M. Aritsugi, Semantic cosine similarity, in: The 7th international student conference on advanced science and technology ICAST, vol. 4, (1) University of Seoul, South Korea, 2012, p. 1.
[20] J. Cohen, A coefficient of agreement for nominal scales, Educ. Psychol. Meas. 20 (1) (1960) 37–46.
[21] Y. Vasiliev, Natural Language Processing with Python and SpaCy: A Practical Introduction, No Starch Press, 2020.
[22] F.C. Rau, A. Wigger, B. Tellhelm, M. Zwick, S. Klumpp, A. Neumann, B. Oltersdorf, K. Amort, K. Failing, M. Kramer, Observer variability and sensitivity of radiographic diagnosis of canine medial coronoid disease, Tierärztliche Prax. Ausg. K: Kleintiere/Heimtiere 39 (05) (2011) 313–322.

[4] https://www.nltk.org/api/nltk.tag.pos_tag.html